

1 The Multidimensional Case

2 Now we consider the case where $\theta = (\theta_1, \theta_2, \dots, \theta_p)$.

3 We define the **Fisher Information matrix** to be the p by p matrix whose (i, j)
4 entry is

$$5 \quad I_{ij}(\theta) = E \left[-\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f(\underline{x}; \theta) \right] = \underline{I}_{ij}(\underline{\theta})$$

6 **The Main Result:** Under appropriate "regularity conditions," if $X_1, X_2, \dots, X_n$
7 are i.i.d. $f(x; \theta)$ then

$$8 \quad \sqrt{n}(\hat{\theta} - \theta_0) \longrightarrow N_p(0, \underline{I}^{-1}(\theta_0))$$

9 Recall that N_p denotes the p -dimensional multivariate normal distribution.

10 **Practical Version:** Under "regularity conditions," $\hat{\theta}$ is approximately nor-
11 mal with variance $\underline{I}^{-1}(\hat{\theta})/n$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$. Derive the joint
2 distribution of the MLE for $\theta = (\mu, \sigma^2)$.
 θ_1, θ_2

$$3 \quad f(x; \underline{\theta}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

$$4 \quad \frac{\partial}{\partial \mu^2} \log f(x; \underline{\theta}) = -\frac{1}{\sigma^2}$$

$$5 \quad \frac{\partial}{\partial \sigma^2} \log f(x; \underline{\theta}) = \frac{1}{2\sigma^4} - \frac{(x-\mu)^2}{\sigma^6}$$

$$6 \quad \frac{\partial^2}{\partial \mu \partial \sigma^2} \log f(x; \underline{\theta}) = -\frac{(x-\mu)}{\sigma^4}$$

So,

$$\left(\frac{X-\mu}{2\sigma^4} \right)$$

$$\underline{I}(\underline{\theta}) = E$$

$$\begin{bmatrix} \frac{1}{\sigma^2} & \left(\frac{X-\mu}{2\sigma^4} \right) \\ \left(\frac{X-\mu}{2\sigma^4} \right) & \frac{(X-\mu)^2}{\sigma^6} - \frac{1}{2\sigma^4} \end{bmatrix}$$

$$\frac{1}{\sigma^2}$$

$$0$$

since

$$E((X-\mu)^2) = \sigma^2$$

$$=$$

$$0$$

$$\frac{1}{2\sigma^4}$$

$$\text{The MLE for } \underline{\theta} \text{ is } \hat{\underline{\theta}} = \begin{bmatrix} \bar{X} \\ \left(\frac{n-1}{n} \right) s^2 \end{bmatrix}$$

It is approx. bivariate normal with covariance

matrix

$$\underline{I}^{-1}(\hat{\underline{\theta}}_0)/n = \begin{bmatrix} \sigma_0^2/n & 0 \\ 0 & 2\sigma_0^4/n \end{bmatrix}$$

In practice, replace σ_0^2 with $\hat{\sigma}^2$, e.g.
approx SE for $\hat{\sigma}^2$ is $\sqrt{\frac{2\hat{\sigma}^4}{n}}$

Recall: $\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2 \approx \text{Normal for } n \text{ large}$

Multivariate Delta Method

We can extend the Delta method to cases where the parameter is multidimensional.

In general, suppose that $\mathbf{Y}_n$ is approximately $N_p(\mu, \Sigma/n)$. Now consider an m -dimensional function $\mathbf{g}(\cdot)$ where $m \leq p$:

$$\mathbf{g}(\mathbf{Y}_n) = \begin{bmatrix} g_1(\mathbf{Y}_n) \\ g_2(\mathbf{Y}_n) \\ \vdots \\ g_m(\mathbf{Y}_n) \end{bmatrix}$$

$\swarrow$ m -dimensional

Then, $\mathbf{g}(\mathbf{Y}_n)$ is approximately $N_m(\mathbf{g}(\mu), \mathbf{G}\Sigma\mathbf{G}^T/n)$ where the (i, j) entry of $\mathbf{G}$ is $\partial g_i(\mathbf{y}) / \partial y_j$. This all assumes that $\mathbf{G}$ is of full rank.

Exercise: Place our parameter estimation problem into the framework of the above statement of the Delta method.

$\mathbf{Y}_n$ is replaced with $\hat{\Theta}_{MLE}$

Σ is replaced with $\mathbf{I}^{-1}(\hat{\Theta}_0)$

μ is replaced with $\hat{\Theta}_0$

Delta method says $\mathbf{g}(\hat{\Theta})$ is approx. normal with mean $\mathbf{g}(\hat{\Theta}_0)$ and covariance matrix

$$\mathbf{G} \mathbf{I}^{-1}(\hat{\Theta}_0) \mathbf{G}^T / n$$

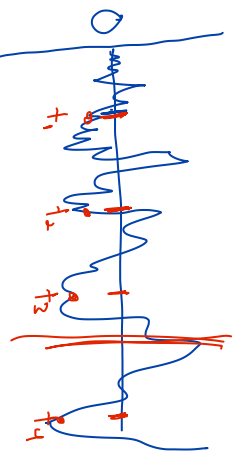
$m \times p$

¹ Case Study: Geometric Brownian Motion

- ² This Part concludes with a simple, but important, example of how careful
- ³ derivation of the MLE and its distribution can lead to useful insight.
- ⁴ We begin by defining some standard concepts regarding Brownian motion.

¹ The Standard Brownian Motion (Wiener Process)

- ² A stochastic process $\{W(t) : t \geq 0\}$ is a **(standard) Brownian motion** if the
- ³ following hold:
 - ⁴ 1. $W(0) = 0$. (almost surely)
 - ⁵ 2. $W(t)$ is continuous as a function of t . (almost surely)
 - ⁶ 3. $W(t_4) - W(t_3)$ and $W(t_2) - W(t_1)$ are independent for $t_1 \leq t_2 \leq t_3 \leq t_4$.
 - ⁷ 4. The distribution of $W(t_2) - W(t_1)$ depends only on $|t_1 - t_2|$ for t_1, t_2 .
 - ⁸ 5. $W(t)$ is Normal($0, t$) for all t .



1 *Brownian Motion with Drift and Scaling*

2 A stochastic process $\{B(t) : t \geq 0\}$ is a **Brownian Motion with drift μ and scaling σ^2** if $\{W(t) : t \geq 0\}$ is a standard Brownian motion and

$$4 \quad B(t) = \underline{\mu t} + \underline{\sigma^2} W(t).$$

5 Note that this implies the following:

- 6 1. $B(0) = 0$. (almost surely)
- 7 2. $B(t)$ is continuous as a function of t . (almost surely)
- 8 3. $B(t_4) - B(t_3)$ and $B(t_2) - B(t_1)$ are independent for $t_1 \leq t_2 \leq t_3 \leq t_4$.
- 9 4. The distribution of $B(t_2) - B(t_1)$ depends only on $|t_1 - t_2|$ for t_1, t_2 .
- 10 5. $B(t)$ is Normal($\mu t, \sigma^2 t$) for all t .

1 *Geometric Brownian Motion*

2 A stochastic process $\{S(t) : t \geq 0\}$ is a **geometric Brownian motion** if $\{B(t) :$
 3 $t \geq 0\}$ is a Brownian motion with drift μ and scaling σ^2 and

$$4 \quad \underline{S(t)} = \underline{S(0)} e^{B(t)}$$

5 where $S(0) > 0$ is the initial value. Define $\underline{\alpha} = \underline{\mu} + \sigma^2/2$.

6 Note that this implies the following:

- 7 1. $S(t)$ is continuous as a function of t . (almost surely)
- 8 **2.** $S(t_4)/S(t_3)$ and $S(t_2)/S(t_1)$ are independent for any $t_1 \leq t_2 \leq t_3 \leq t_4$.
- 9 3. The distribution of $S(t_2)/S(t_1)$ depends only on $|t_1 - t_2|$ for any t_1, t_2 .
- 10 4. $S(t)$ is Lognormal($\log(S(0)) + \mu t, \sigma^2 t$) for all t .
- 11 5. $S(t)$ has expected value $S(0)e^{\alpha t}$. $\alpha > 0 \Rightarrow$ price increasing
- 12 **6.** $\log(S(t_2)/S(t_1))$ is Normal($\mu(t_2 - t_1), \sigma^2(t_2 - t_1)$) for all $t_1 \leq t_2$. $\alpha < 0 \Rightarrow$ price decreasing

- 1 Let's consider the geometric Brownian motion model. Can we use data
- 2 (observed prices) to estimate α and σ^2 ?
- 3 Suppose we observe the stock price over the period $[0, T]$. We will model
- 4 the price as a geometric Brownian motion $\{S(t) : t \geq 0\}$.
- 5 Divide the time interval into n equal-width subintervals, with $\Delta = T/n$.
- 6 Define, for $i = 1, 2, \dots, n$,
- 7
$$Y_i = \log \left(\frac{S(t_{i-1} + \Delta)}{S(t_{i-1})} \right)$$
- 8 where $t_i = (i/n)T$, the times at which the price is observed.
- 9 From Property 6 of a geometric Brownian motion, $Y_1, Y_2, \dots, Y_n$ are i.i.d.
- 10 $N(\Delta\mu, \Delta\sigma^2)$.

- 1 **Exercise:** Find the MLE of (α, σ^2) . Approximate the distribution of $\hat{\theta}_{MLE}$ this pair.

2 The $Y_1, Y_2, \dots, Y_n$ are i.i.d. $N(\Delta\mu, \Delta\sigma^2)$

3 Let $\theta_1 = \Delta\mu$ and $\theta_2 = \Delta\sigma^2$ "original parameters"

$$\theta = (\theta_1, \theta_2)$$

5 We know from previous derivations that $\hat{\theta}_{MLE}$

6 is approx. normal with covariance

$$I_n^{-1}(\hat{\theta})/n = \begin{bmatrix} \hat{\theta}_2/n & 0 \\ 0 & 2\hat{\theta}_2^2/n \end{bmatrix}, \quad \theta_2 = V(Y_i)$$

But, want the covariance matrix for the MLE of (α, σ^2) .

In the notation of the Delta method,

$$\alpha = g_1(\underline{\theta}) = \mu + \frac{\sigma^2}{2} = \frac{\theta_1}{\Delta} + \frac{\theta_2}{2\Delta}$$

$$\sigma^2 = g_2(\underline{\theta}) = \frac{\theta_2}{\Delta}$$

By invariance prop. of MLE

$$\hat{\alpha}_{MLE} = \frac{\hat{\theta}_1}{\Delta} + \frac{\hat{\theta}_2}{2\Delta} \quad \hat{\theta}_1 = \bar{Y}$$

$$\hat{\sigma}_{MLE}^2 = \frac{\hat{\theta}_2}{\Delta} \quad \hat{\theta}_2 = \left(\frac{n-1}{n}\right) s_Y^2$$

The matrix G in this case:

$$G = \begin{bmatrix} \frac{\partial g_1(\underline{\theta})}{\partial \theta_1} & \frac{\partial g_1(\underline{\theta})}{\partial \theta_2} \\ \frac{\partial g_2(\underline{\theta})}{\partial \theta_1} & \frac{\partial g_2(\underline{\theta})}{\partial \theta_2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\Delta} & \frac{1}{2\Delta} \\ 0 & \frac{1}{\Delta} \end{bmatrix}$$

$$S_b, \quad \hat{G}(\hat{I}'^{-1}(\theta_0)/n) \hat{G}^T$$

$$= \begin{bmatrix} \frac{\theta_0^2}{n\Delta^2} + \frac{\theta_0^2}{2n\Delta^2} & \theta_0^2/n\Delta^2 \\ \frac{\theta_0^2}{n\Delta^2} & 2\theta_0^2/n\Delta^2 \end{bmatrix}$$

note $n\Delta = T$
 $\sigma^2 = \theta_0/\Delta$

$$= \begin{bmatrix} \sigma^2/T + \frac{\sigma^4}{2n} & \sigma^4/n \\ \sigma^4/n & 2\sigma^4/n \end{bmatrix} \quad E(S(t)) = S(t)e^{\alpha t}$$

Exercise: What does this result suggest regarding how large n should be chosen for the best estimation of σ^2 ? What about for α ?

$$V(\hat{\sigma}^2) = 2\sigma^4/n \Rightarrow \text{in order to estimate } \sigma^2 \text{ well, choose } n \text{ larger}$$

But, the estimation of α is limited by T
since $V(\hat{\alpha}) = \sigma^2/T + \sigma^4/2n$. Need to increase T to drive $V(\hat{\alpha})$ to 0.

Typical values of σ around 0.2. So, if want $SE(\hat{\alpha}) \leq 0.01$, ie. $\sqrt{\sigma^2/T + \sigma^4/2n} \leq 0.01$
need $T \gtrsim 400$ years

- ¹ Is it, in fact, a good idea to sample more frequently in order to estimate
- ² σ^2 ? See Ait-Sahalia, Mykland, and Zhang (2005), “How Often to Sam-
- ³ ple a Continuous-Time Process in the Presence of Market Microstructure
- ⁴ Noise,” *The Review of Financial Studies*, Volume 18.